Supplementary Evaluation Report
Point-Based Mask Refinement for X-Ray Instance Segmentation

1 Scope

This report collects every evaluation produced for the ten models trained on the PIXray official split: complete COCO box and mask metrics, precision, recall and F1, per-class results, detection error analysis, mask quality, computational performance, and training losses. It accompanies the main manuscript and uses the same verified prediction files. All models were trained once for 60 epochs with seed 42 and evaluated at their final epoch on the 1,486 test images (4,508 instances, 15 classes). Table 1 lists the models.

Table 1: Methods evaluated in this report. Short names are used in all later tables. Parameters in millions (total / trainable). All methods use the official PIXray split, 60 epochs and seed 42.

Short name	Method	Key difference	Total	Train.
Mask R-CNN	Mask R-CNN (ResNet-101 + FPN)	single-stage mask head	63.0	62.8
Cascade FPN	Cascade Mask R-CNN (ResNet-101 + FPN)	3-stage cascade	96.1	95.8
Parent (AttFPN)	Cascade Mask R-CNN + AttFPN (parent)	attention-weighted pyramid	97.1	96.9
Recall RPN	Cascade + AttFPN + Recall RPN	wider anchors, larger proposal budget	97.1	96.9
Bilayer	Cascade + Bilayer Mask Head	recall RPN + bilayer head, aux. occluder masks	105.2	104.9
PIDray R0/R1	PIDray R0/R1 Adaptation	image-level R0/R1 gate	98.7	98.5
RSPrompter	RSPrompter Adaptation (SAM)	SAM ViT-L, LoRA, learned prompts	332.7	27.6
Occ	Occ Adaptation (SAM)	SAM bilayer decoder, aux. occluder masks	336.7	31.6
PR-Transformer	Point Refinement–Transformer (ours)	shared point refiner, 2 attention blocks	97.6	97.3
PR-MLP	Point Refinement–MLP (ours)	shared point refiner, 2 point-wise MLP blocks	97.6	97.3

2 Metric Definitions

- **AP** is COCO average precision averaged over IoU thresholds 0.50:0.05:0.95 and all classes; AP_{50} and AP_{75} use one threshold. AP_S, AP_M and AP_L restrict ground truth to small, medium and large objects using the annotation area field, which stores box area in PIXray.
- **AR**$_n$ is COCO average recall with at most n detections per image; $AR_S/AR_M/AR_L$ use at most 100.
- **Precision, recall, F1.** Detections scoring at least a threshold are matched to ground truth with COCO's own rule (class-wise, one-to-one, in descending score order) at IoU ≥ 0.50 or ≥ 0.75. Precision = TP/(TP+FP), recall = TP/(number of objects), F1 = 2PR/(P+R). *Micro* values pool all classes; *macro* F1 averages per-class F1. We report the operating point at score 0.50 and the best micro F1 over thresholds 0.05–0.95.
- **Detection errors** classify each false positive at score ≥ 0.50 by its box IoU with ground truth (duplicate, wrong class, poor localization, other, background); unmatched objects are *missed*.
- **Mask quality** compares matched masks at original resolution: IoU, Dice = $2|P \cap G|/(|P| + |G|)$, and boundary IoU on a band of 2% of the image diagonal.

3 COCO Detection and Segmentation Metrics

Tables 2 and 3 give all twelve COCO numbers. The MLP point-refinement variant reaches the highest mask AP, 71.50%, against 69.12% for its parent. Table 4 and Figure 1 break mask AP down by IoU threshold.

4 Precision, Recall and F1

Tables 5 and 6 report precision, recall and F1. At the lenient IoU ≥ 0.50 the models are close: the best mask F1 at score 0.50 is 94.77% (Parent (AttFPN)), and the MLP variant reaches 94.66% (P 96.89%, R 92.52%). At the strict IoU

Table 2: Bounding-box COCO metrics on the PIXray test split (%, higher is better; bold: best). AP averages IoU 0.50:0.05:0.95; AR_n allows at most n detections per image; S/M/L group ground truth by the annotation area field (box area in PIXray).

Method	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AR_1	AR_{10}	AR_{100}	AR_S	AR_M	AR_L
Mask R-CNN	75.49	94.79	86.65	51.53	66.66	74.46	74.71	79.66	79.66	54.01	71.30	78.19
Cascade FPN	80.02	94.77	88.94	61.28	71.41	78.65	77.56	82.75	82.75	63.93	74.46	81.12
Parent (AttFPN)	80.21	94.52	89.17	61.05	71.61	78.82	**77.79**	82.92	82.92	63.53	74.55	81.23
Recall RPN	**80.34**	**95.29**	89.38	65.92	71.89	78.93	77.73	**83.08**	**83.08**	**68.76**	75.20	81.49
Bilayer	80.23	94.93	**89.71**	62.01	**71.99**	78.63	77.71	82.99	82.99	64.49	**75.24**	81.02
PIDray R0/R1	80.12	94.92	89.48	**65.95**	71.59	**79.90**	77.70	82.88	82.88	68.40	74.82	**82.38**
RSPrompter	73.84	95.08	85.37	37.92	65.85	72.10	72.48	77.63	77.63	42.91	70.84	75.27
Occ	73.70	95.18	85.34	40.69	65.71	71.92	72.45	77.48	77.48	45.65	70.59	75.13
PR-Transformer	80.01	94.69	89.35	58.78	71.72	78.56	77.48	82.73	82.73	61.31	74.85	80.98
PR-MLP	80.00	94.85	89.16	64.31	71.30	78.61	77.49	82.68	82.68	66.83	74.53	80.94

Table 3: Instance mask COCO metrics on the PIXray test split (%, higher is better; bold: best). AP averages IoU 0.50:0.05:0.95; AR_n allows at most n detections per image; S/M/L group ground truth by the annotation area field (box area in PIXray).

Method	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AR_1	AR_{10}	AR_{100}	AR_S	AR_M	AR_L
Mask R-CNN	66.42	92.94	75.72	36.25	52.50	67.29	66.20	70.62	70.62	57.96	65.14	68.17
Cascade FPN	68.79	93.32	78.57	37.27	56.52	68.20	67.36	71.89	71.89	55.48	65.89	68.89
Parent (AttFPN)	69.12	93.53	79.30	38.70	57.23	68.45	67.74	72.23	72.23	58.72	66.68	69.14
Recall RPN	68.94	93.81	79.00	39.98	56.80	69.35	67.44	72.13	72.13	60.62	66.56	69.98
Bilayer	69.22	93.80	79.13	37.37	56.82	68.45	67.72	72.41	72.41	59.54	67.14	69.09
PIDray R0/R1	68.99	93.39	79.14	41.25	56.58	69.22	67.60	72.14	72.14	62.97	66.20	69.86
RSPrompter	61.24	91.52	69.14	30.26	45.33	63.53	60.47	64.88	64.88	40.36	56.54	64.34
Occ	61.43	91.80	69.07	19.97	45.44	63.66	60.67	65.06	65.06	40.68	56.56	64.23
PR-Transformer	70.07	93.32	78.62	42.02	57.31	70.44	68.56	73.16	73.16	59.38	67.04	71.11
PR-MLP	**71.50**	**93.92**	**81.41**	**44.90**	**58.80**	**72.55**	**69.78**	**74.42**	**74.42**	**63.23**	**67.52**	**73.15**

Table 4: Mask AP (%) at each IoU threshold (bold: best).

Method	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95
Mask R-CNN	92.94	91.61	90.11	86.66	82.31	75.72	65.45	51.14	26.92	1.34
Cascade FPN	93.32	92.59	90.91	88.54	84.51	78.57	69.54	55.55	32.16	2.17
Parent (AttFPN)	93.53	92.69	90.98	88.38	84.52	79.30	70.18	55.75	33.35	2.55
Recall RPN	93.81	92.97	91.35	88.64	84.10	79.00	69.52	55.62	32.20	2.17
Bilayer	93.80	93.08	91.36	89.00	84.70	79.13	70.08	55.31	33.53	2.22
PIDray R0/R1	93.39	92.79	91.13	88.84	84.18	79.14	69.70	55.46	32.86	2.40
RSPrompter	91.52	89.75	86.22	82.21	77.04	69.14	56.60	41.04	18.54	0.37
Occ	91.80	89.78	85.91	82.11	77.16	69.07	57.60	41.45	18.82	0.58
PR-Transformer	93.32	92.61	90.90	88.21	84.71	78.62	70.69	57.88	36.50	7.28
PR-MLP	**93.92**	**93.28**	**91.94**	**89.46**	**85.97**	**81.41**	**73.52**	**60.02**	**37.98**	**7.55**

≥ 0.75, where mask quality matters, the best mask F1 is 85.01% (PR-MLP); the MLP variant scores 85.01% against 83.77% for the parent (+1.24 pp) and 83.19% for the transformer variant. At IoU ≥ 0.75 the MLP variant also has the highest mask precision and recall (P 87.02%, R 83.10%, against 85.70% and 81.92% for the parent). The SAM-based Occ model has the highest box recall (94.23%) but the lowest precision (box 88.44%, mask 86.09%). Figures 2 and 3 show the full precision–recall curves and F1 against the score threshold.

5 Per-Class Results

Tables 7–10 give mask AP, box AP and mask F1 for every class. The MLP variant has the best mask AP in 12 of 15 classes and the best strict (IoU ≥ 0.75) mask F1 in 6 classes.

Table 5: Mask precision (P), recall (R) and F1 (%) with COCO one-to-one, class-wise, score-ordered matching at score threshold 0.50, plus the best micro F1 over thresholds 0.05–0.95 and the threshold where it occurs. Micro averages pool all classes; macro F1 averages per-class F1. Bold: best.

	IoU ≥ 0.50						IoU ≥ 0.75				
Method	P	R	F1	macro F1	best F1	at score	P	R	F1	best F1	at score
Mask R-CNN	94.06	92.37	93.21	93.28	93.99	0.76	81.43	79.97	80.69	82.91	0.95
Cascade FPN	96.52	92.28	94.35	94.33	94.59	0.72	84.90	81.17	82.99	84.25	0.95
Parent (AttFPN)	**96.96**	92.68	**94.77**	**94.73**	94.83	0.54	85.70	81.92	83.77	84.73	0.95
Recall RPN	95.90	**92.79**	94.32	94.29	94.61	0.77	84.11	81.39	82.73	84.25	0.95
Bilayer	96.29	92.72	94.47	94.47	94.80	0.61	84.93	81.79	83.33	84.41	0.93
PIDray R0/R1	96.65	92.72	94.65	94.62	94.73	0.68	85.36	81.90	83.60	84.73	0.94
RSPrompter	86.51	91.59	88.98	89.08	91.97	0.91	68.41	72.43	70.36	75.16	0.95
Occ	86.09	91.73	88.82	88.95	92.01	0.87	68.17	72.63	70.33	75.17	0.95
PR-Transformer	96.58	92.66	94.58	94.55	**94.84**	0.66	84.95	81.50	83.19	84.13	0.83
PR-MLP	96.89	92.52	94.66	94.63	94.83	0.63	**87.02**	**83.10**	**85.01**	**85.89**	0.91

Table 6: Box precision (P), recall (R) and F1 (%) with COCO one-to-one, class-wise, score-ordered matching at score threshold 0.50, plus the best micro F1 over thresholds 0.05–0.95 and the threshold where it occurs. Micro averages pool all classes; macro F1 averages per-class F1. Bold: best.

	IoU ≥ 0.50						IoU ≥ 0.75				
Method	P	R	F1	macro F1	best F1	at score	P	R	F1	best F1	at score
Mask R-CNN	95.28	93.57	94.42	94.48	94.97	0.75	89.29	87.69	88.48	90.07	0.91
Cascade FPN	97.49	93.21	95.31	95.30	95.49	0.67	92.95	88.86	90.86	91.53	0.86
Parent (AttFPN)	**97.75**	93.43	95.54	95.52	95.58	0.49	**93.48**	89.35	91.37	91.75	0.88
Recall RPN	96.88	93.74	95.29	95.28	95.44	0.75	92.41	89.42	90.89	91.75	0.85
Bilayer	97.10	93.50	95.27	95.28	95.50	0.59	93.11	**89.66**	91.35	91.95	0.85
PIDray R0/R1	97.62	93.66	**95.60**	**95.59**	**95.65**	0.56	93.46	**89.66**	**91.52**	91.96	0.78
RSPrompter	88.98	94.21	91.52	91.60	93.88	0.82	81.94	86.76	84.28	87.99	0.91
Occ	88.44	**94.23**	91.25	91.37	93.99	0.79	81.22	86.54	83.79	87.96	0.95
PR-Transformer	97.48	93.52	95.46	95.44	95.60	0.66	93.32	89.53	91.38	**92.01**	0.83
PR-MLP	97.44	93.06	95.20	95.18	95.38	0.62	93.19	89.00	91.05	91.72	0.76

Table 7: Per-class mask AP (%; bold: best per class). Classes sorted from hardest to easiest by mean mask AP. MRCNN = Mask R-CNN; CFPN = Cascade FPN; Parent = Cascade + AttFPN; RRPN = Recall RPN; Bilay. = Bilayer head; RSP = RSPrompter; PR-T / PR-M = our point refinement with transformer / MLP blocks.

Class	MRCNN	CFPN	Parent	RRPN	Bilay.	PIDray	RSP	Occ	PR-T	PR-M
Dart	35.44	35.83	35.95	36.52	36.80	35.59	20.20	19.41	36.25	**36.87**
Saw_blade	41.20	44.61	44.91	44.11	45.27	45.05	36.87	37.11	42.89	**51.04**
Knife	51.32	54.09	55.16	54.86	54.59	54.85	43.71	44.25	54.78	**56.80**
Screwdriver	53.71	59.02	58.95	59.36	59.81	58.92	50.49	50.70	61.32	**62.11**
Pliers	55.77	59.99	59.84	58.88	59.96	60.09	49.92	49.69	59.83	**64.07**
Lighter	64.52	66.54	67.51	66.96	67.44	67.64	56.57	56.65	67.74	**68.33**
Razor_blade	68.50	68.79	68.65	**69.93**	68.86	68.81	60.13	59.95	68.43	69.64
Wrench	65.19	69.37	70.33	69.68	69.46	70.20	64.87	65.70	70.94	**73.01**
Scissors	72.09	74.81	74.43	73.94	74.58	74.15	60.66	61.27	75.86	**76.69**
Fireworks	73.39	74.99	75.92	74.79	75.97	75.11	70.96	72.08	**77.77**	76.70
Battery	77.56	79.09	79.33	**80.07**	79.14	78.90	72.86	73.35	79.83	79.74
Gun	81.66	82.79	84.04	83.05	83.83	83.24	79.36	79.30	84.93	**85.52**
Hammer	83.41	84.93	84.90	84.50	85.10	84.73	78.61	78.05	85.12	**85.98**
Pressure_vessel	84.97	87.74	88.10	88.06	87.91	88.16	84.39	85.04	90.75	**91.17**
Bat	87.57	89.22	88.84	89.36	89.63	89.43	89.04	88.87	94.65	**94.89**
Best in class	0	0	0	2	0	0	0	0	1	12

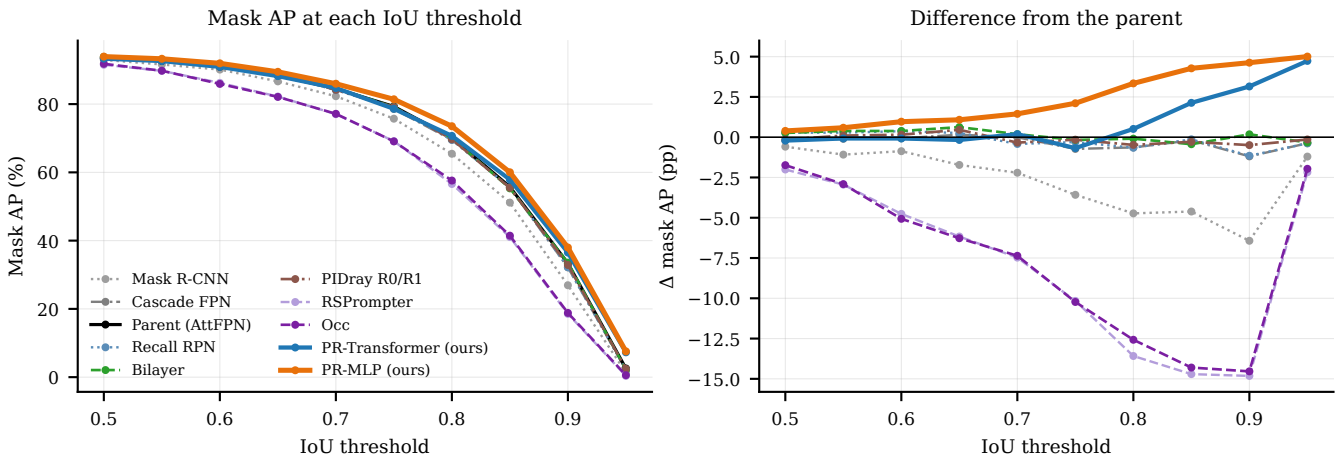


Figure 1: Mask AP at each IoU threshold (left) and its difference from the parent model (right).

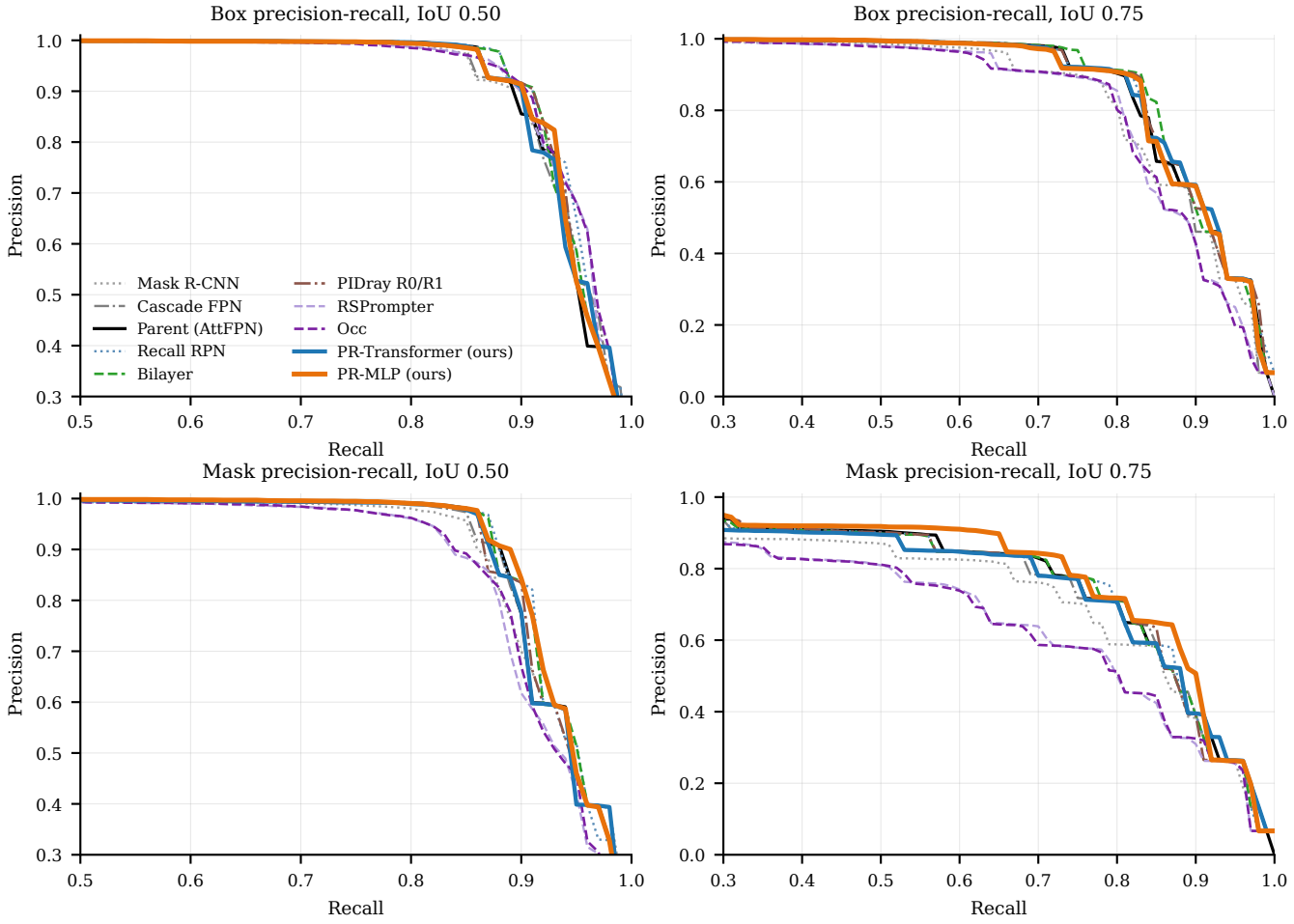


Figure 2: Class-averaged precision–recall curves for boxes (top) and masks (bottom) at IoU 0.50 (left) and 0.75 (right).

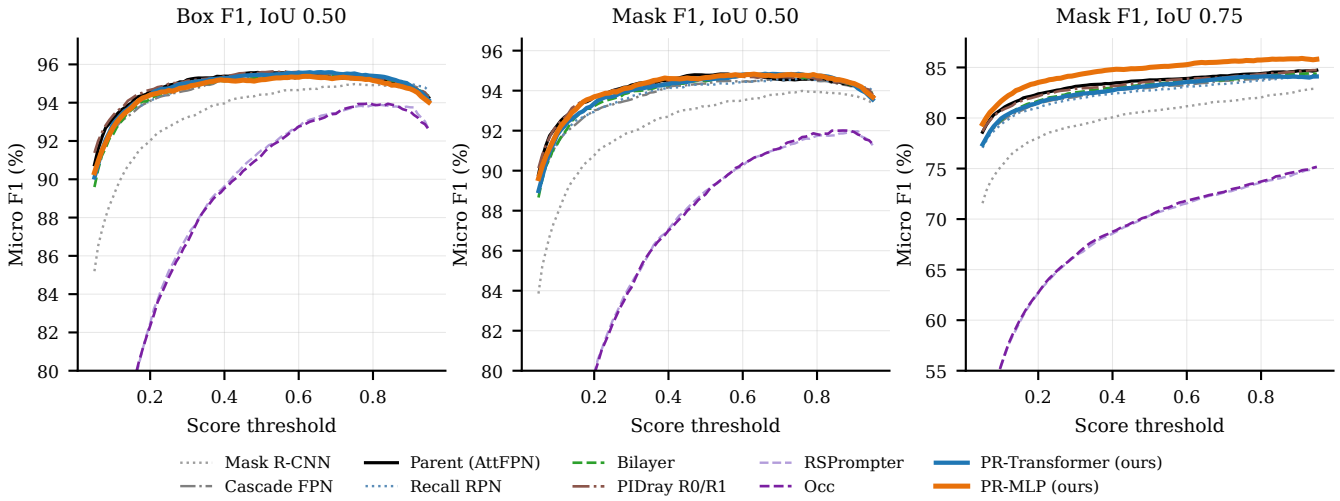


Figure 3: Micro F1 against the score threshold.

Table 8: Per-class box AP (%; bold: best per class). Classes sorted from hardest to easiest by mean mask AP. MRCNN = Mask R-CNN; CFPN = Cascade FPN; Parent = Cascade + AttFPN; RRPN = Recall RPN; Bilay. = Bilayer head; RSP = RSPrompter; PR-T / PR-M = our point refinement with transformer / MLP blocks.

Class	MRCNN	CFPN	Parent	RRPN	Bilay.	PIDray	RSP	Occ	PR-T	PR-M
Dart	54.92	59.63	60.65	61.04	**61.32**	60.75	54.14	53.60	59.57	59.64
Saw_blade	68.89	74.89	74.04	74.99	75.10	75.23	69.64	69.38	**75.24**	73.48
Knife	65.63	71.50	72.01	72.70	**73.35**	72.34	65.63	65.18	71.31	71.86
Screwdriver	70.61	77.07	**77.88**	77.30	77.54	77.54	70.55	70.33	76.42	76.22
Pliers	83.70	87.73	87.65	**88.30**	87.41	88.19	79.59	78.90	86.90	87.52
Lighter	66.92	72.55	72.17	72.26	72.48	72.02	66.29	66.12	71.97	**72.67**
Razor_blade	67.85	70.76	70.31	**71.12**	70.49	70.64	67.92	67.99	70.79	70.77
Wrench	78.52	84.42	**85.02**	85.00	84.63	84.70	77.07	76.68	83.97	84.46
Scissors	79.43	82.96	82.50	83.10	82.51	82.60	75.41	75.26	83.24	**84.10**
Fireworks	72.51	74.80	75.52	75.04	75.52	74.05	70.27	71.25	**76.19**	75.03
Battery	74.98	78.16	78.73	**79.26**	78.06	78.15	76.58	77.13	78.80	78.34
Gun	85.69	88.06	**89.01**	87.80	87.85	88.52	81.22	81.79	88.09	88.92
Hammer	85.80	89.37	**90.17**	89.50	89.85	89.47	83.00	82.23	90.13	89.53
Pressure_vessel	84.92	**91.49**	91.22	91.17	90.95	90.83	82.39	81.53	91.35	90.69
Bat	91.99	**96.86**	96.29	96.57	96.45	96.76	87.95	88.07	96.16	96.74
Best in class	0	2	4	3	2	0	0	0	2	2

Table 9: Per-class mask F1 (%) at IoU ≥ 0.50 and score 0.50 (bold: best per class). Rows shown with three decimals would tie at two. Classes sorted from hardest to easiest by mean mask AP. MRCNN = Mask R-CNN; CFPN = Cascade FPN; Parent = Cascade + AttFPN; RRPN = Recall RPN; Bilay. = Bilayer head; RSP = RSPrompter; PR-T / PR-M = our point refinement with transformer / MLP blocks.

Class	MRCNN	CFPN	Parent	RRPN	Bilay.	PIDray	RSP	Occ	PR-T	PR-M
Dart	84.86	87.21	87.38	87.80	86.93	**88.15**	77.18	77.48	87.23	85.95
Saw_blade	84.35	87.83	88.73	86.06	87.87	86.88	78.48	78.15	87.17	**89.52**
Knife	84.08	88.40	89.67	89.91	89.34	88.79	79.04	79.33	**90.05**	89.87
Screwdriver	86.56	90.52	91.40	90.94	91.40	90.88	84.85	85.62	**91.61**	90.97
Pliers	97.35	96.38	**97.53**	97.14	**97.53**	**97.53**	91.88	90.91	96.59	97.15
Lighter	93.77	93.52	93.10	92.82	93.36	**94.09**	86.62	85.71	93.00	93.89
Razor_blade	92.10	92.82	92.25	91.68	92.43	**93.41**	86.13	85.35	92.72	91.78
Wrench	93.28	95.22	**95.68**	95.09	94.51	94.95	91.04	89.60	94.12	94.81
Scissors	**95.62**	94.55	94.68	94.06	94.55	94.35	90.60	90.50	95.05	95.58
Fireworks	**98.57**	98.21	98.38	98.03	98.38	98.38	93.38	95.10	**98.57**	98.03
Battery	94.72	95.11	95.98	95.16	94.99	**96.56**	92.05	92.31	95.98	95.68
Gun	97.14	98.11	98.46	98.63	98.29	97.95	93.01	93.14	98.46	**99.32**
Hammer	98.911	99.220	99.532	99.222	99.532	99.532	98.009	98.457	**99.533**	99.532
Pressure_vessel	98.00	97.98	**98.48**	98.32	98.32	97.83	95.36	93.98	98.32	97.48
Bat	99.83	99.83	99.66	99.48	99.66	**100.00**	98.63	98.63	99.83	99.83
Best in class	2	0	3	0	1	6	0	0	4	2

Table 10: Per-class mask F1 (%) at the strict IoU ≥ 0.75, score 0.50 (bold: best per class). Rows shown with three decimals would tie at two. Classes sorted from hardest to easiest by mean mask AP. MRCNN = Mask R-CNN; CFPN = Cascade FPN; Parent = Cascade + AttFPN; RRPN = Recall RPN; Bilay. = Bilayer head; RSP = RSPrompter; PR-T / PR-M = our point refinement with transformer / MLP blocks.

Class	MRCNN	CFPN	Parent	RRPN	Bilay.	PIDray	RSP	Occ	PR-T	PR-M
Dart	30.95	33.00	33.55	31.54	32.35	**34.06**	1.23	0.90	31.18	32.78
Saw_blade	50.34	53.26	58.45	56.79	57.29	57.80	31.96	32.85	53.08	**66.43**
Knife	65.77	70.53	73.13	71.29	72.41	71.34	48.44	48.27	72.67	**75.63**
Screwdriver	71.55	76.21	77.42	**79.22**	78.85	77.64	60.94	60.54	78.83	78.70
Pliers	75.76	83.05	81.59	79.24	82.35	84.25	58.67	59.00	79.92	**86.91**
Lighter	80.93	**85.27**	83.23	81.55	84.38	84.65	69.52	66.67	82.49	84.81
Razor_blade	89.40	**89.71**	89.53	88.97	88.54	89.15	77.84	76.31	88.89	88.72
Wrench	85.84	87.70	89.05	88.15	87.86	88.02	82.07	82.39	88.38	**89.91**
Scissors	90.02	91.04	91.13	89.86	90.69	90.81	77.95	80.14	90.81	**92.04**
Fireworks	92.47	92.11	93.00	91.56	92.61	90.84	83.62	84.62	**94.27**	93.00
Battery	90.44	91.95	93.39	92.59	91.85	92.84	88.22	89.56	93.10	**93.66**
Gun	93.78	96.40	**97.09**	96.58	95.90	95.56	90.73	91.18	97.08	96.92
Hammer	97.978	97.972	98.284	97.356	98.284	98.908	95.253	95.988	**98.911**	97.972
Pressure_vessel	95.00	95.29	96.46	95.97	**96.64**	96.49	94.04	92.68	96.30	96.13
Bat	99.83	99.83	99.31	99.48	99.66	**100.00**	98.63	98.63	99.83	99.83
Best in class	0	2	1	1	1	2	0	0	2	6

6 Detection Error Analysis

Table 11 and Figure 4 classify every detection at score ≥ 0.50. Refinement does not improve detection: the MLP variant makes 110 false positives and misses 313 objects, the most of any model, against 97 and 296 for the parent. The SAM-based Occ model makes the most false positives (555). Figure 5 shows the class confusion matrices of the parent and the MLP variant.

Table 11: Detection outcomes at score ≥ 0.50 using box IoU (counts over 4,508 test instances). Duplicate: extra detection of an already-matched object; wrong class: IoU ≥ 0.5 with an object of another class; poor loc.: same class but $0.1 \leq$ IoU < 0.5; other: another class with $0.1 \leq$ IoU < 0.5; background: IoU < 0.1 with every object. Bold: best (most TP; fewest errors).

Method	TP ↑	Dup. ↓	Wrong cls. ↓	Poor loc. ↓	Other ↓	Backgr. ↓	All FP ↓	Missed ↓
Mask R-CNN	4218	8	100	28	36	37	209	290
Cascade FPN	4202	4	57	12	13	22	108	306
Parent (AttFPN)	4212	2	**51**	11	**9**	24	**97**	296
Recall RPN	4226	4	58	19	21	34	136	282
Bilayer	4215	5	57	21	16	27	126	293
PIDray R0/R1	4222	5	54	10	13	**21**	103	286
RSPrompter	4247	24	151	92	115	144	526	261
Occ	**4248**	30	167	86	120	152	555	**260**
PR-Transformer	4216	4	**51**	**9**	15	30	109	292
PR-MLP	4195	**1**	52	14	12	31	110	313

7 Mask Quality

On matched instances (Table 12) the MLP variant has the highest mean mask IoU, 0.8688 against 0.8573 for the parent, and 49.23% of its matched masks reach IoU ≥ 0.90, against 43.97% for the parent.

8 Computational Performance

Table 13 and Figure 6 give model size and logged timings. The MLP variant runs at 28.7 ms per image against 25.6 ms for the parent and 30.8 ms for the transformer variant, and trained in 9.06 GPU-hours against 6.37. These timings come from runs on different days and indicate scale only.

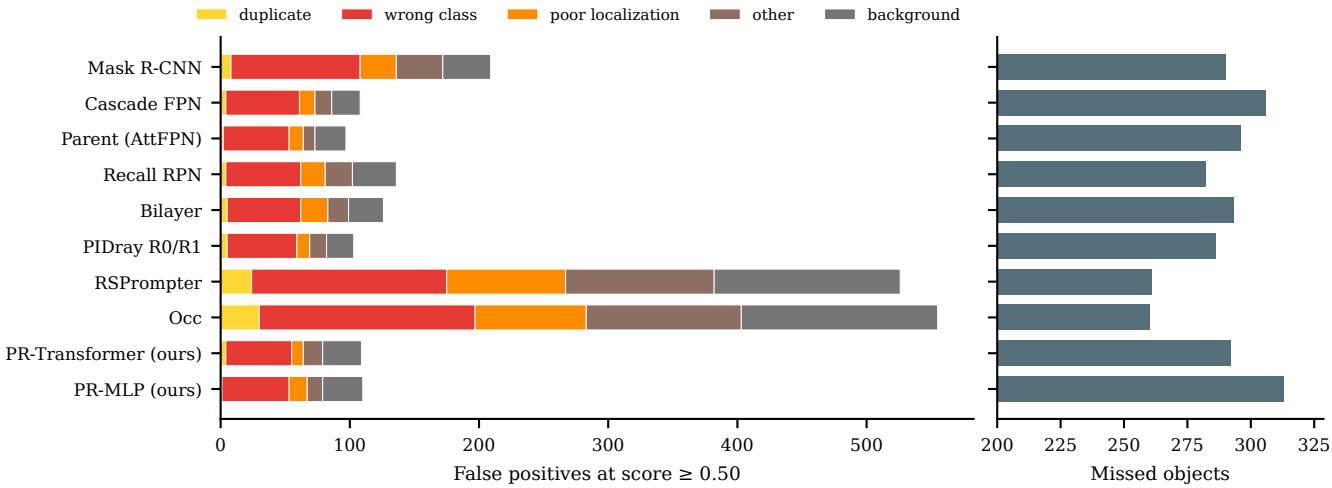


Figure 4: False-positive types (left) and missed objects (right) at score ≥ 0.50.

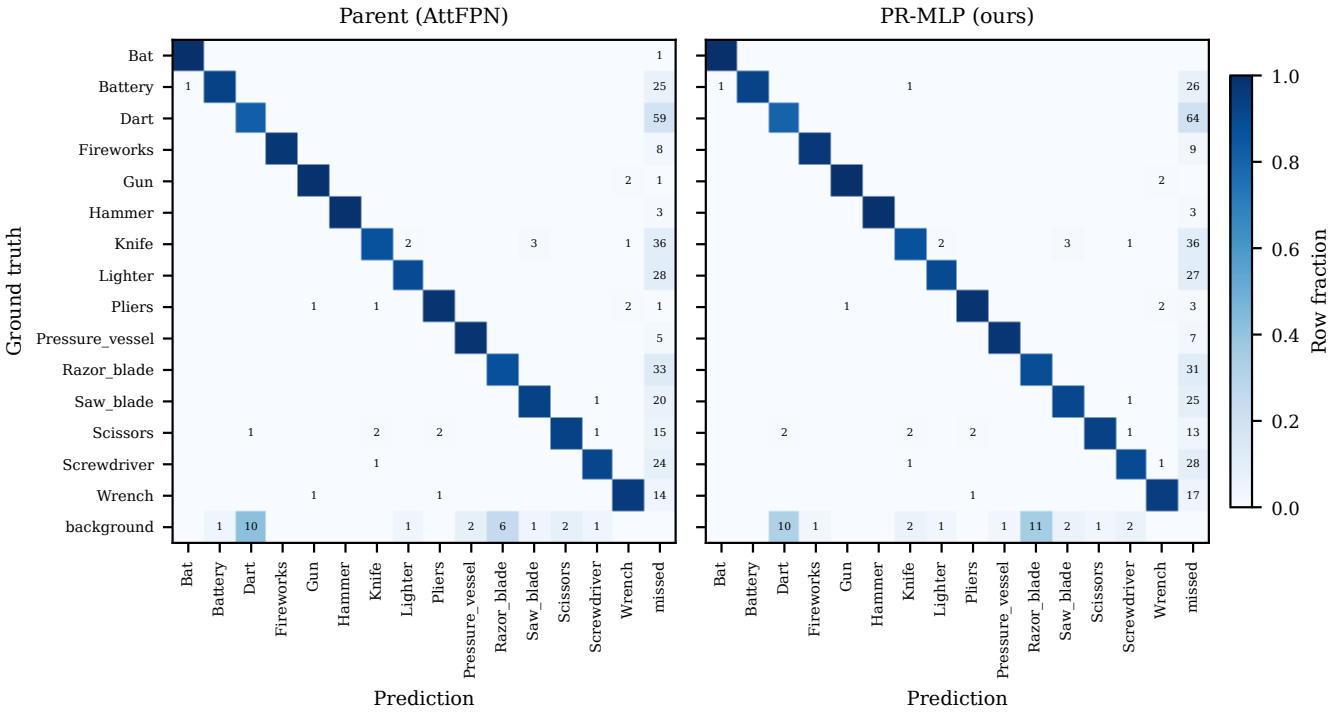


Figure 5: Row-normalised confusion matrices at score ≥ 0.50 (box IoU ≥ 0.5). Rows are ground-truth classes plus 'background' (false positives on no object); columns are predicted classes plus 'missed'. Off-diagonal counts are printed.

Table 12: Mask quality on matched instances (score ≥ 0.50, one-to-one, same class, box IoU ≥ 0.50), at original resolution. Coverage: % of all ground-truth instances matched. Dice $= 2|P \cap G|/(|P| + |G|)$; boundary IoU uses a band of 2% of the image diagonal. Unmatched: predictions at score ≥ 0.50 that match no object (lower is better). Bold: best.

Method	Coverage	Mean IoU	Median IoU	IoU≥0.75	IoU≥0.90	Dice	Boundary IoU	Unmatched ↓
Mask R-CNN	93.57	0.8468	0.8835	85.44	40.09	0.9129	0.8351	209
Cascade FPN	93.21	0.8547	0.8878	87.08	43.19	0.9181	0.8435	108
Parent (AttFPN)	93.43	0.8573	0.8902	87.68	43.97	0.9196	0.8465	**97**
Recall RPN	93.74	0.8539	0.8870	86.82	43.07	0.9174	0.8427	136
Bilayer	93.50	0.8565	0.8888	87.45	44.22	0.9191	0.8456	126
PIDray R0/R1	93.66	0.8558	0.8892	87.40	43.68	0.9185	0.8448	103
RSPrompter	94.21	0.8112	0.8503	76.83	26.23	0.8893	0.7974	526
Occ	**94.25**	0.8121	0.8505	76.96	26.45	0.8898	0.7984	554
PR-Transformer	93.52	0.8612	0.8948	87.12	47.18	0.9217	0.8513	109
PR-MLP	93.06	**0.8688**	**0.8990**	**89.27**	**49.23**	**0.9267**	**0.8590**	110

Table 13: Model size, training cost and inference speed. Training and test figures are as logged on single NVIDIA RTX 4090 runs made on different days, so small differences are not meaningful; no values are bolded. GPU-h: training time; s/iter and Train GB: mean training iteration time and peak training memory; ms/img, FPS and Test GB: logged inference time, throughput and peak memory, averaged over all logged evaluation runs (Runs). The first logged window of each run is excluded.

Method	Params (M)	Train. (M)	GPU-h	s/iter	Train GB	ms/img	FPS	Test GB	Runs
Mask R-CNN	63.0	62.8	3.45	0.112	2.00	18.9	52.9	0.43	1
Cascade FPN	96.1	95.8	5.22	0.170	3.04	24.7	40.6	0.55	1
Parent (AttFPN)	97.1	96.9	6.37	0.208	3.44	25.6	39.0	0.69	1
Recall RPN	97.1	96.9	6.70	0.219	3.47	29.8	33.5	0.69	1
Bilayer	105.2	104.9	8.45	0.279	4.36	29.9	33.4	0.72	1
PIDray R0/R1	98.7	98.5	6.64	0.218	3.50	24.9	40.2	0.70	1
RSPrompter	332.7	27.6	7.35	0.239	9.76	51.1	19.6	1.59	3
Occ	336.7	31.6	9.55	0.313	12.64	54.6	18.3	1.68	3
PR-Transformer	97.6	97.3	9.34	0.309	7.36	30.8	32.4	0.70	1
PR-MLP	97.6	97.3	9.06	0.299	7.17	28.7	34.9	0.70	1

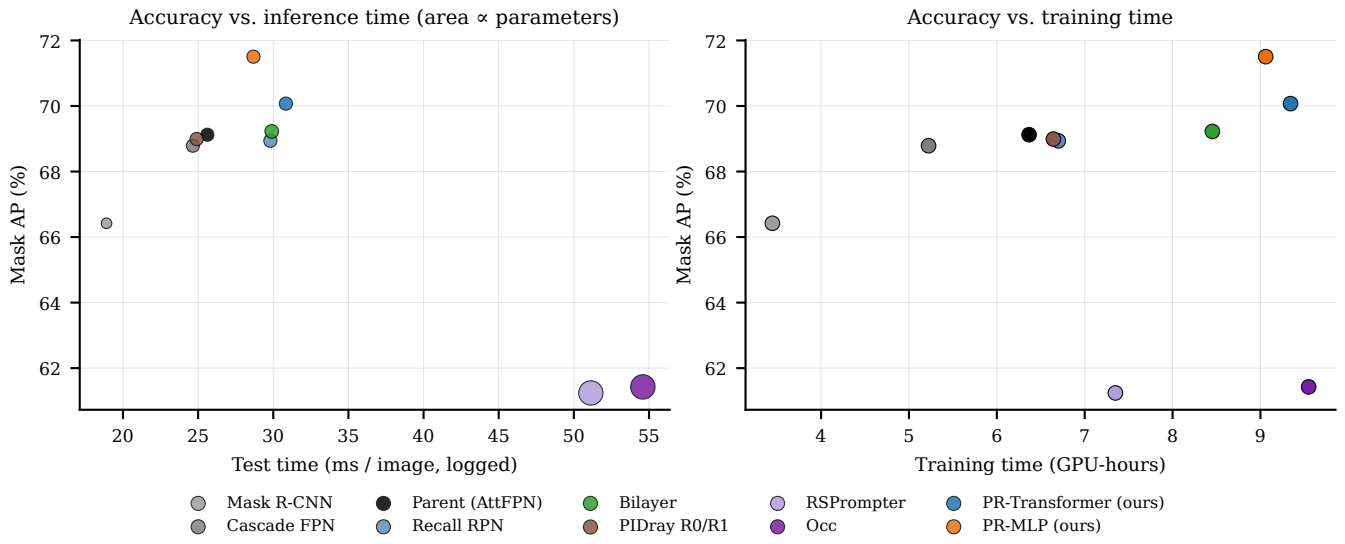


Figure 6: Mask AP against logged inference time (marker area proportional to total parameters) and against training time.

9 Training Losses

Table 14 lists each loss term at the first and final epoch; Table 15 details the point-refinement terms per cascade stage; Figure 7 shows the training curves.

Table 14: Training losses, mean over logged iterations of the first and final (60th) epoch. Terms are summed over cascade stages as logged, i.e. with stage weights 1 / 0.5 / 0.25 applied; '–' marks a term the model does not have. Totals include different terms per model and are not comparable across models.

Method	Total (ep. 1)	Total (ep. 60)	RPN cls	RPN box	cls	box	mask	occluder mask	R0 gate	point
Mask R-CNN	0.9639	0.1816	0.0029	0.0092	0.0281	0.0649	0.0765	–	–	–
Cascade FPN	1.3859	0.2310	0.0038	0.0043	0.0318	0.0532	0.1379	–	–	–
Parent (AttFPN)	1.3592	0.2216	0.0038	0.0041	0.0294	0.0489	0.1354	–	–	–
Recall RPN	1.5918	0.2495	0.0033	0.0051	0.0361	0.0660	0.1390	–	–	–
Bilayer	1.6410	0.2692	0.0033	0.0052	0.0369	0.0674	0.1333	0.0231	–	–
PIDray R0/R1	1.3466	0.2230	0.0039	0.0042	0.0294	0.0495	0.1358	–	0.0001	–
RSPrompter	0.8876	0.1628	0.0058	0.0015	0.0986	0.0546	0.0022	–	–	–
Occ	0.9156	0.1658	0.0082	0.0015	0.0985	0.0548	0.0022	0.0006	–	–
PR-Transformer	2.4024	0.6365	0.0048	0.0049	0.0340	0.0546	0.1279	–	–	0.4102
PR-MLP	2.4094	0.6409	0.0047	0.0049	0.0349	0.0551	0.1287	–	–	0.4126

Table 15: Point-refinement training terms per cascade stage (final-epoch means). Point loss is the stage-weighted mean of the two level losses; BCE l_0/l_1 are the unweighted binary cross-entropies at the 56^2 and 112^2 levels; target mean is the fraction of sampled points inside the object; points is the number of sampled points per region.

Model	Stage	Mask loss	Point loss	BCE l_0	BCE l_1	Target mean l_0	Target mean l_1	Points
PR-Transformer	1	0.0707	0.2238	0.2203	0.2230	0.4854	0.4878	196
	2	0.0380	0.1234	0.2395	0.2411	0.4983	0.4999	196
	3	0.0192	0.0630	0.2418	0.2443	0.4994	0.5007	196
PR-MLP	1	0.0714	0.2261	0.2217	0.2240	0.4788	0.4800	196
	2	0.0382	0.1236	0.2394	0.2409	0.4914	0.4917	196
	3	0.0191	0.0629	0.2422	0.2446	0.4944	0.4942	196

10 Evaluator Variant and Verification

PIXray's test annotations include an object with id 0, which the official pycocotools matcher confuses with "unmatched". Table 16 shows that correcting this changes AP by at most 0.07 pp; all AP tables use the official evaluator, and precision, recall and F1 use the corrected matching. Every prediction file matched its recorded SHA-256 hash, every recomputed box and mask AP equalled the audited value exactly, and the mask-quality values for the earlier models reproduced previous computations exactly. The Occ and RSPrompter models carry incompletely recorded prediction provenance, as noted in the main manuscript.

Table 16: Official pycocotools AP versus an evaluator that correctly handles the ground-truth annotation with id 0 (%). All reported tables use the official evaluator; precision, recall and F1 use the corrected matching.

	Box AP			Mask AP		
Method	official	corrected	Δ	official	corrected	Δ
Mask R-CNN	75.491	75.529	+0.038	66.421	66.443	+0.022
Cascade FPN	80.016	80.056	+0.039	68.787	68.825	+0.038
Parent (AttFPN)	80.211	80.244	+0.033	69.123	69.162	+0.039
Recall RPN	80.343	80.392	+0.050	68.938	68.977	+0.039
Bilayer	80.233	80.281	+0.048	69.223	69.270	+0.047
PIDray R0/R1	80.120	80.152	+0.032	68.991	69.030	+0.040
RSPrompter	73.843	73.883	+0.040	61.243	61.266	+0.023
Occ	73.697	73.747	+0.050	61.428	61.461	+0.033
PR-Transformer	80.010	80.064	+0.054	70.073	70.138	+0.065
PR-MLP	79.999	80.041	+0.042	71.505	71.536	+0.032

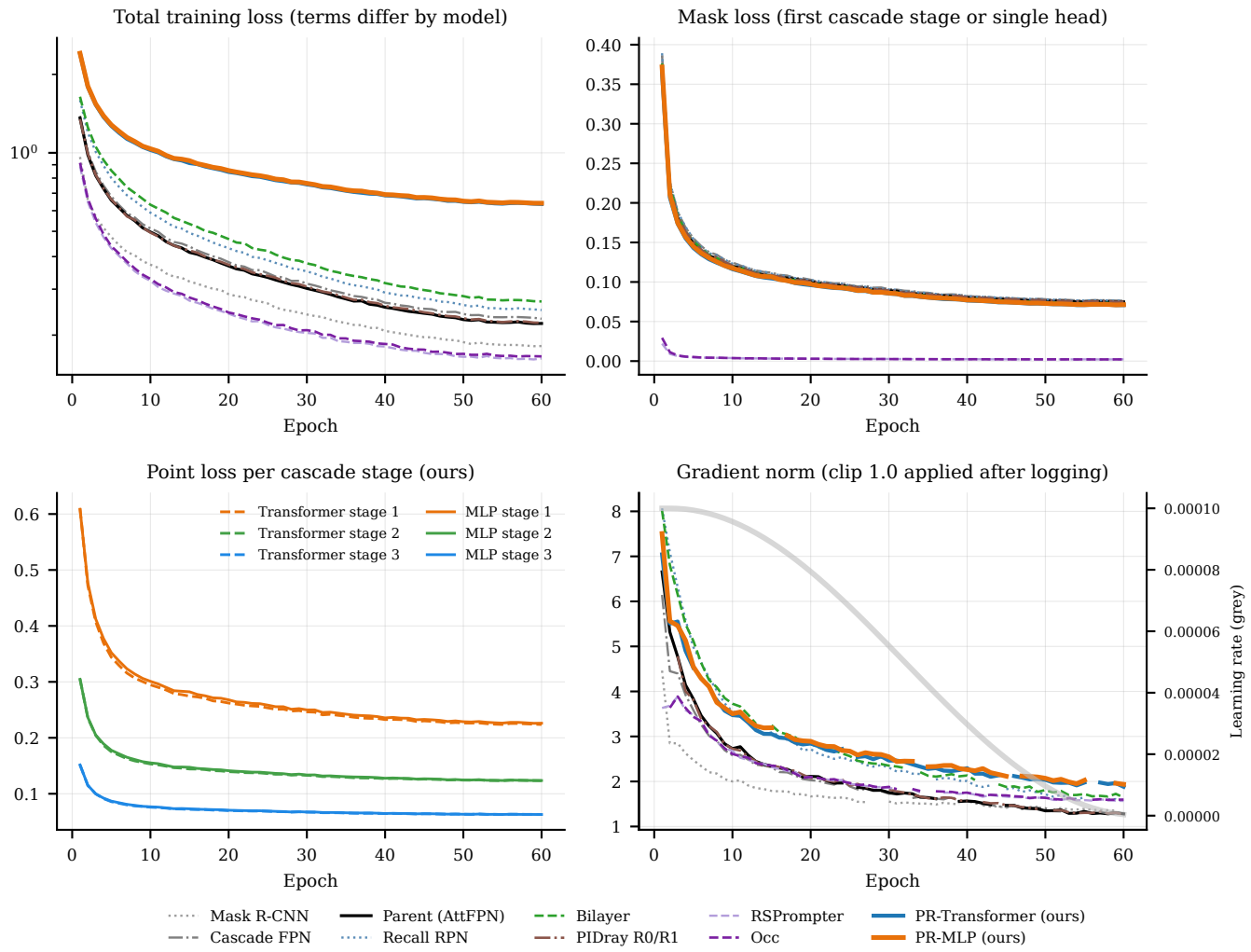


Figure 7: Per-epoch means of the total loss, the first-stage mask loss, our point loss per stage, and the gradient norm with the learning-rate schedule.